

ZOMATO: What Happens After You Tap 'Order Now'



Think About Your Last Order



How did Zomato know which restaurants to show YOU — not your roommate, not your colleague, but specifically YOU?



How did they promise '35 minutes' before the food was even cooked and no rider was assigned?



How did they pick YOUR rider from 480,000 delivery partners across India?

By the end of tonight, you'll know exactly how.

The Scale of Operations

1,200

Orders Every Single Minute



20 orders per second

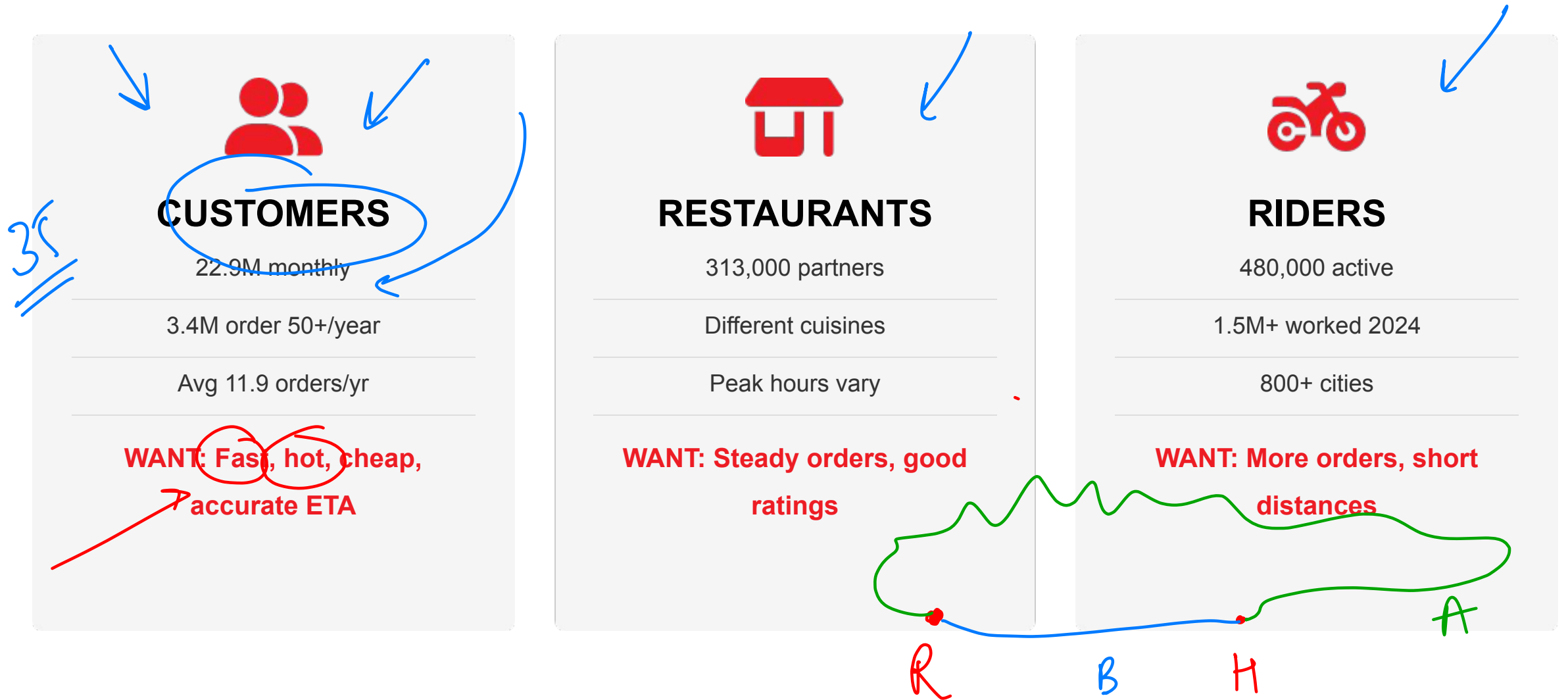
647 million orders in FY23

3+ billion delivered all-time

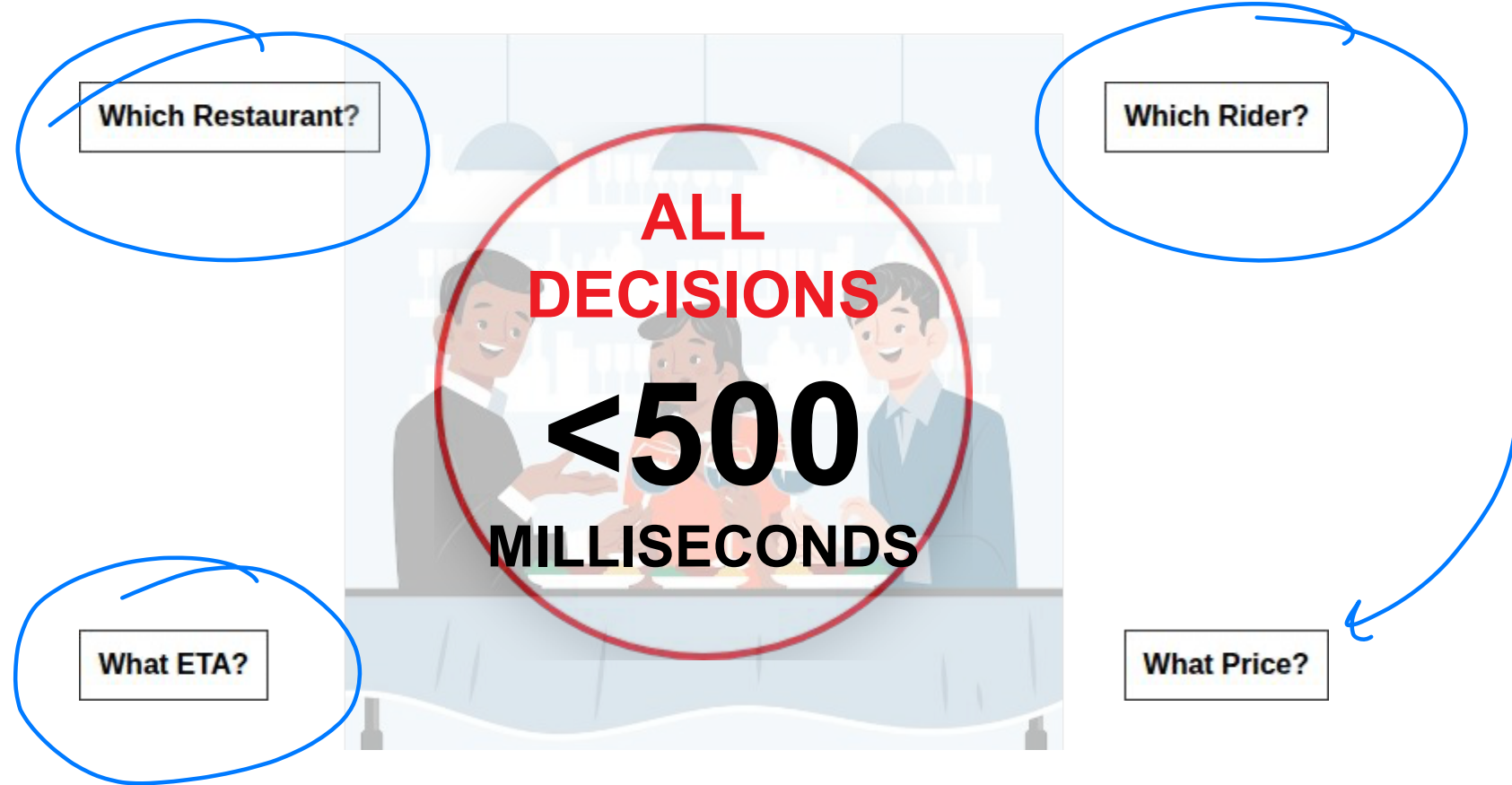
144,000 orders during this class

Source: Zomato Investor Deck FY24

Three Groups. Three Goals. One Order.



Every Order = Real-Time Coordination



A human blink takes 300-400ms. You cannot blink faster than Zomato thinks.

What Zomato Must Do For EVERY Order

1. Show the 'right' 20 restaurants to THIS specific user
2. Predict delivery time before food is cooked or rider assigned
3. Pick optimal rider from 480,000 options
4. Adjust prices based on real-time supply/demand
5. Handle fraud, refunds, complaints automatically

This is why Machine Learning exists — some problems are HUMANLY IMPOSSIBLE.

500 ms
Half a second for ALL decisions

1 person: ~1 order/min

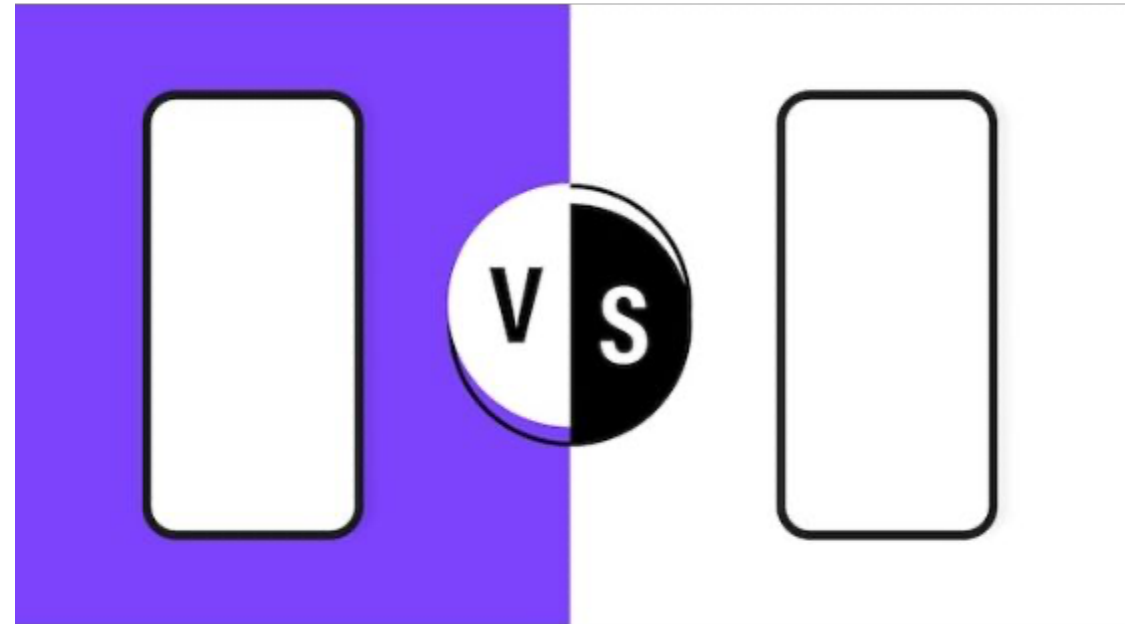
100 people: ~100 orders/min

1,000 people: Still can't match 1,200/min at 500ms

PART 1

When You Open the App

Why Your Zomato Looks Different
From Everyone Else's



Same App. Same City. Same Time. Different Results.

The image shows two side-by-side smartphone screens representing a food delivery app. The left screen is for 'USER A' and the right for 'USER B'. Both screens show a list of restaurants with their names, cuisines, and estimated delivery times. Each restaurant entry has a small square icon to its left and a blue checkmark to its right. A large red '≠' symbol is placed between the two screens. Handwritten blue notes are present: 'fast food' with a slash next to it is written next to User A's screen, and 'healthy option' is written next to User B's screen. An arrow points from the 'USER B' label to the top of the right screen.

User	Restaurant	Cuisine	Time	Handwritten Note
USER A	Pizza Hut	Italian	30 mins	fast food
	Behrouz Biryani	Biryani	45 mins	
	Chinese Wok	Chinese	25 mins	
	Dominos	Pizza	30 mins	
	McDonald's	Burger	20 mins	
USER B	Salad Days	Healthy	35 mins	healthy option
	Healthy Bites	Salad	40 mins	
	South Indian Kitchen	Dosa	25 mins	
	Subway	Sandwich	20 mins	
	Freshii	Bowls	30 mins	
	(Unlabeled)			

There is no single 'Zomato'. There are 22.9 million different Zomatos.

The Ranking Formula

Every Restaurant Gets a Personalized Score — Recalculated Every Time You Open the App

$$\text{Score} = f(\text{User Profile}, \text{Restaurant Status}, \text{Context})$$

YOUR PROFILE

- Complete order history
- Cuisines browsed
- Price range patterns
- Viewed but not ordered

RESTAURANT

- Current rating (live)
- Current prep time
- Current order load
- Active promotions

CONTEXT

- Time of day / Week
- Weather
- Distance from you
- Events (IPL, holidays)
- Rider availability

Why This Investment Pays Off

Better ML = Better Business

21%

Orders placed faster when
personalization works

+₹3

GMV per app open

+₹6

Average Order Value increase

"These are direct results from Zomato's ranking improvements. The ML model's quality directly impacts revenue."

Rank

$R_1 - 9$
 $R_2 - 7$
 $R_3 - 1$
 $R_4 - 10$

Filter 1: Your City/Region 15,000

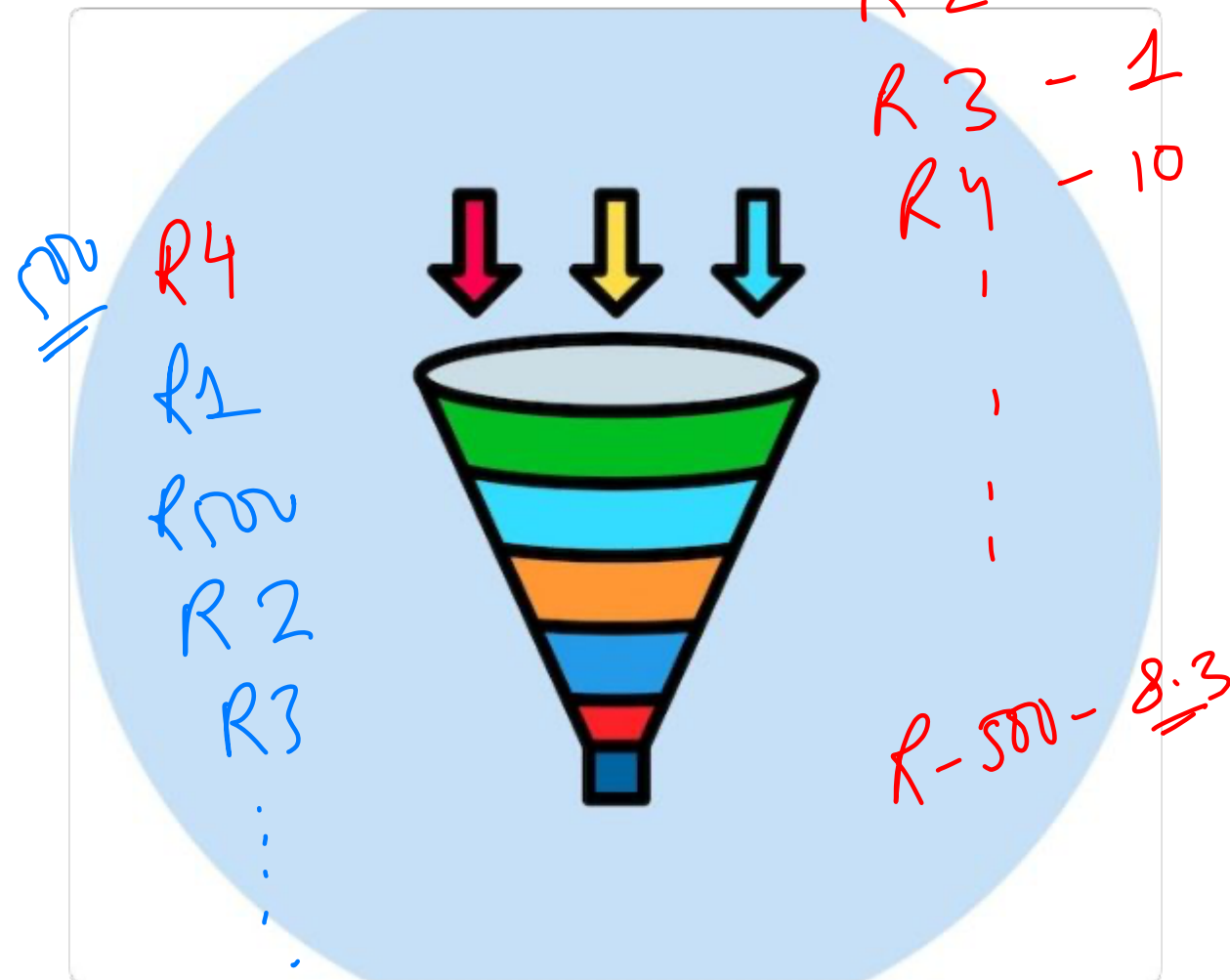
Filter 2: Delivery Radius 2,000

Filter 3: Open & Available 500

Filter 4: Personalized Score 50

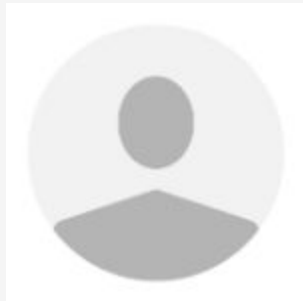
Filter 5: Display Limit 20

99.994% eliminated. The 20 you see are SPECIFICALLY chosen for YOU.
Someone in your same building sees a different 20.



| The Cold Start Problem

Day 1: The Problem



- ✗ Zero order history
- ✗ Zero browsing behavior
- ✗ Zero preferences

"Yet they must show relevant restaurants — bad first impression = uninstall"

The Solution: Proxy Data

- **Location:** Neighborhood implies income & price sensitivity.
- **Phone Model:** Premium phone often indicates higher spending power.
- **Time:** Browsing at 1 PM vs 8 PM (Lunch vs Dinner).
- **Collaborative Filtering:** "People like you in this area also liked..."

Result: Even Day 1 gets reasonable recommendations. Every tap after that — system learns more about YOU.

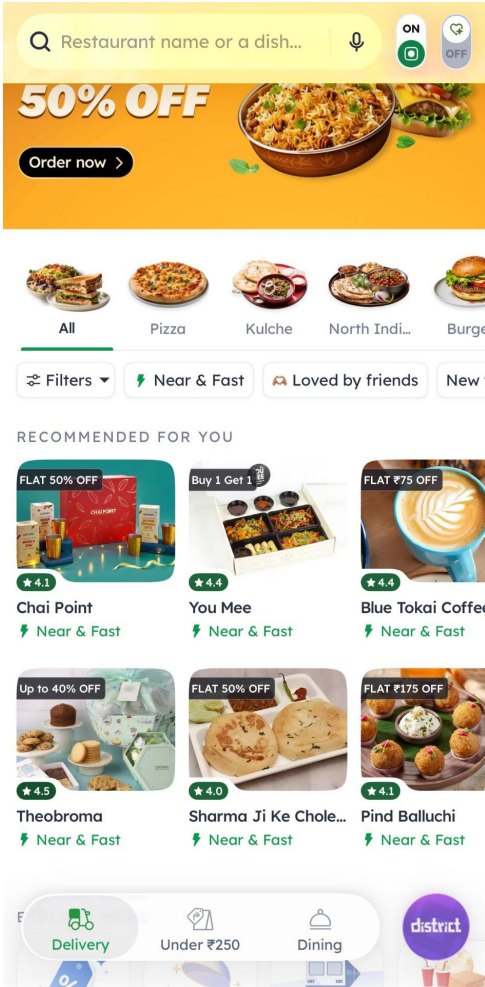
PART 2

The ETA Promise

How Zomato Predicts the Future



'Delivered in 35 Minutes' — A Future Promise



9:00

35 min

9:00

Unknown

9:00

35 + 45
9:35

The Complexity Behind the Number

Zomato commits to a **SPECIFIC** number. Not 'around 35'. Not '30-45'. Exactly 35.

⚠ Food has **NOT** been cooked yet. ✓

⚠ No rider has been assigned. ✓

⚠ Traffic 30 min from now is **UNKNOWN**. ✓

⚠ Restaurant might get a sudden rush. ✓

| Decomposing the 35 Minutes

$$\text{ETA} = \text{Food Prep Time} + \text{Travel Time} + \text{Buffer}$$



Food Prep

15 min

Kitchen processing time



Travel

12 min

Rider to Rest. + Rest. to You



Buffer

+8 min

Safety margin for uncertainty

Each component has its own ML model. Three predictions combined into one number.

Predicting Food Preparation Time

Insight: Zomato does NOT ask the restaurant. Restaurants are either optimistic or conservative. Zomato **PREDICTS** based on data.

Input Features

- **Dish Type:** Biryani vs Sandwich.
- **Order Size:** Number of items.
- **Current Load:** How busy is the kitchen right now?
- **History:** Historical performance of this restaurant.
- **Time:** Day of week, Time of day.

Result: 9% improvement in 5-minute accuracy = 100+ more accurate predictions/min.

The Model: Bidirectional LSTM



Uses **Word2Vec embeddings** to understand context:

- 'Hyderabadi Biryani' $\approx$ 'Lucknowi Biryani' (Similar time)
- 'Chicken Biryani' $\neq$ 'Chicken Sandwich' (Different time)

Predicting FUTURE Traffic

Challenge: Need to know traffic 20-30 minutes from NOW. Zomato uses **480,000 Moving GPS Sensors** (Riders) to know exactly how long every road takes.

Road Condition / Scenario	Travel Time
Road X (7 PM Weekday)	8 min
Road X (7 PM Weekday + Rain)	14 min
Road X (7 PM + IPL Match Nearby)	25 min

Model: LightGBM (Gradient Boosting)

Result: 9% fewer ETA breaches.

100 → X mins → X + Y min
9/10/20

| Every Failure Makes It Smarter

The Feedback Loop

Unlike humans, ML systems never forget a lesson. Every correction stays in the model forever.

Example:

1. Orders to Building X consistently 8 min late.
2. Investigation: Slow elevator.
3. Update: Add 8 min buffer for Building X.
4. Next prediction: Accurate.

ETA + 8 min



5 Minute Break



🚶 Stretch your legs

🚰 Grab some water

🍔 Order food & **WATCH THE ETA**

Up Next: How Zomato picks YOUR rider from 480,000 options.

PART 3

Matching Your Rider

The Optimization Challenge



The Assignment Problem

You Ordered. ETA Promised. Now Zomato Has
SECONDS to Assign a Rider.

Live Snapshot:

- **15 riders** within 2 km radius
- **8** are free right now
- **3** are finishing nearby orders
- **4** are heading opposite direction

Question: Which one gets YOUR order?



Why Nearest Isn't Always Best

Your Instinct: Pick the Nearest Rider. But That's Often Wrong.

❌ Scenario A: The Nearest

Rider A: 500m away

BUT: Currently picking up another order.

AND: That order goes in the opposite direction from you.

Result: Your delivery delayed by 15+ min.

✅ Scenario B: The Optimal

Rider B: 2 km away

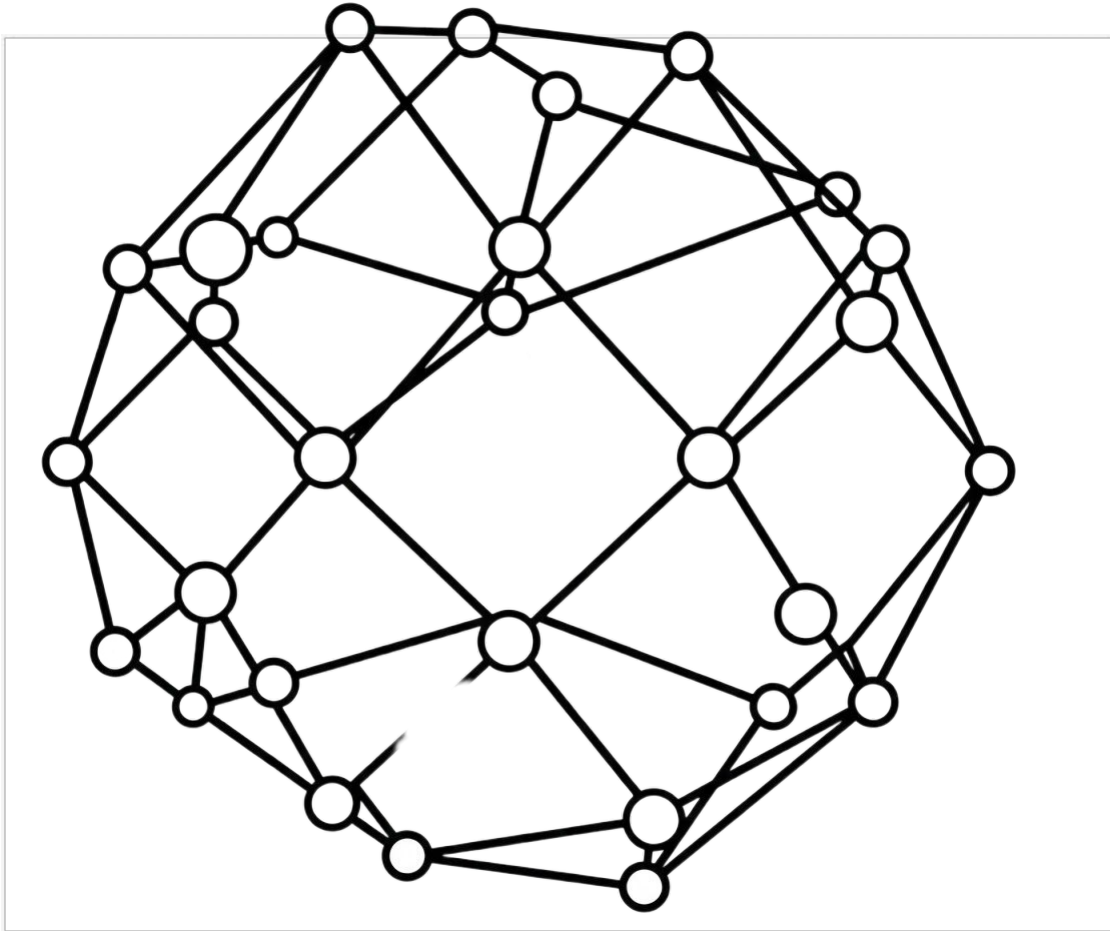
BUT: Heading toward your restaurant already.

AND: Has no current active orders.

Result: Faster actual delivery time.

Insight: Individual optimal $\neq$ System optimal. Zomato must optimize for ALL 1,200 orders/minute, not just yours.

System-Level Optimization



Optimizing for the ENTIRE System

The Problem: If everyone gets the nearest rider, popular areas accumulate riders while quiet areas starve. Someone in Zone B could wait 30 mins because all riders migrated to Zone A.

The Solution: Combinatorial Optimization

Minimize TOTAL system delay, not just individual delay.

- Every active order considered simultaneously
- Rider location, direction, and load factored in
- Future demand predicted and pre-positioned

Order Batching Logic

One Rider, Two Orders

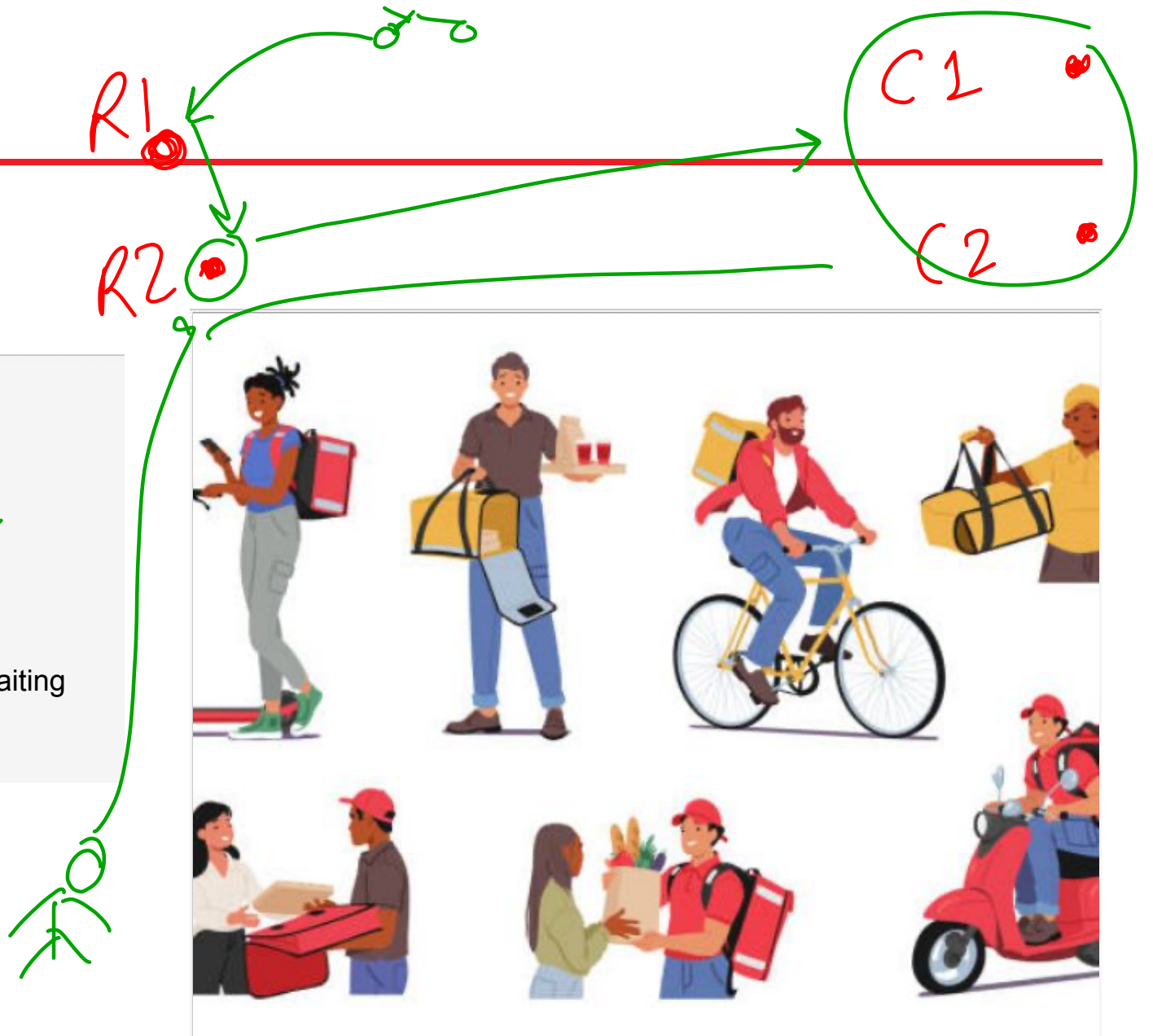
Why your rider sometimes picks up another order first.

The Logic:

1. Restaurant A is near Restaurant B. ✓
2. Your home is near the other customer's home. ✓
3. Your order is "delayed" 3 min... but it's still faster than waiting for a totally different rider. ✓

System Benefits:

- Increases rider earnings per trip ✓
- Reduces overall customer wait times ✓
- Maximizes fleet efficiency ✓



The AI Behind Rider Assignment



R1 R2 R3 R4 . . . R10

Reactive Dispatch

Model: Deep Q-Network (DQN) ✓

The same type of AI that learns to play chess and video games.

- Tries different assignment strategies
- Measures outcomes in real-time
- Learns what works through trial & error
- Multi-agent: considers all riders simultaneously

Active Dispatch

Concept: Proactive Positioning ✓

Recommending where IDLE riders should go before orders happen.

- If Zone A is about to get busy, send riders BEFORE the rush
- Predicting demand before orders come in
- Heat maps guide rider positioning

PART 4

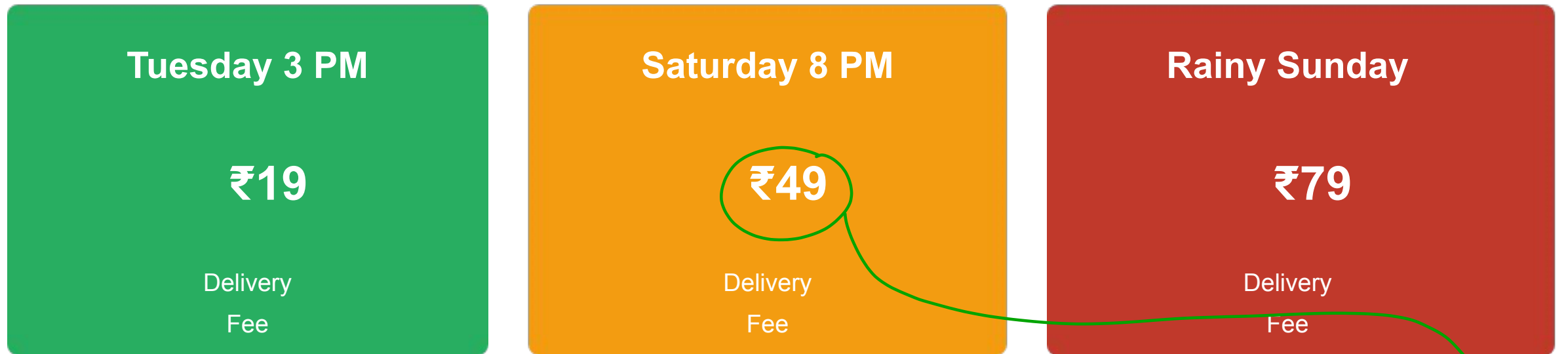
Why Prices Change

Supply and Demand in
Real-Time



The Price Mystery

Same Restaurant. Same Distance. Three Different Prices.



Why? Is Zomato just being greedy?

Answer: Partially... but there's real economics at play.

Demand ↑
Supply ↔

Supply and Demand Economics



Economics 101 Implemented by Algorithms

High Demand (Saturday 8 PM):

Everyone wants to order. Demand is HIGH. Supply is CONSTANT. Price RISES.

The Effects:

- **Effect 1:** Some customers think '₹49? I'll cook instead.' → Demand decreases.
- **Effect 2:** Riders see higher fees, more come online → Supply increases.

*"Price is the balancing mechanism between supply and demand.
Calculated in real-time by algorithms."*

Hyperlocal Surge Pricing

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Same City. Same Time. Different Prices.

Real-time pricing engine responding to supply/demand in each micro-zone.

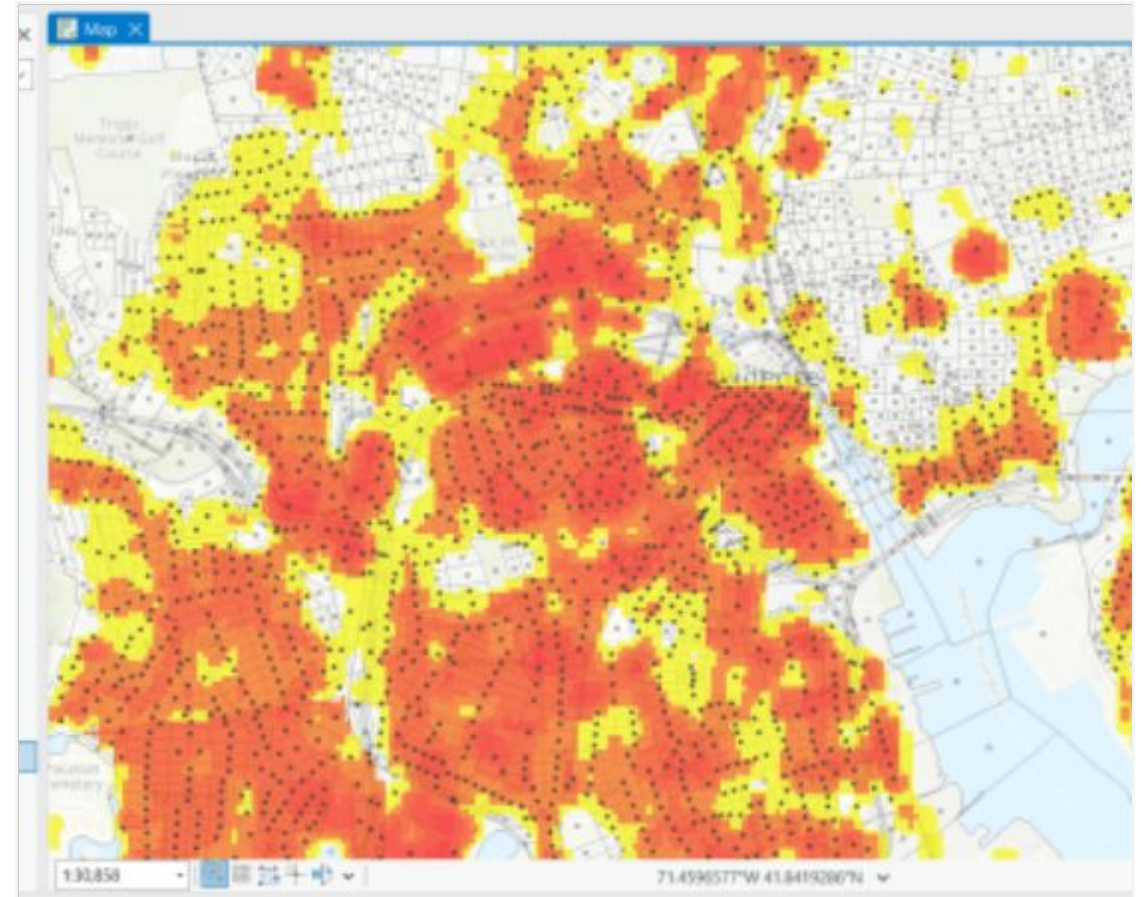
✓ **Zone A:** 1.5x surge (Orange) ✓

Zone B: 1.0x / No surge (Green)

Zone C: 1.8x surge (Red)

"Your neighbor 3 km away might have no surge while you're paying 80% extra."

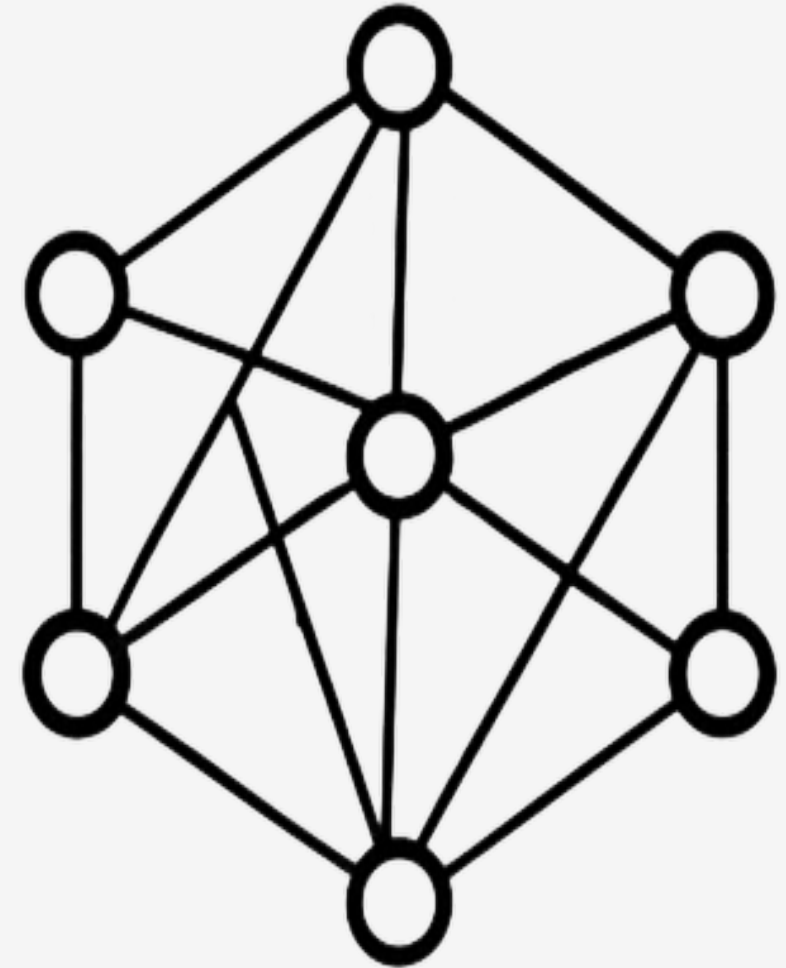
*Recalculates every few SECONDS.



PART 5

The Data Behind It All

Everything Is Tracked



Every Tap Is Data — The System Sees Everything

Direct Actions

- Every order placed (items, time, amount)
- Restaurants viewed and for how long
- Items added to cart but NOT ordered
- Scroll patterns and browsing behavior
- Which promotions clicked

Indirect Signals

- Time between opening app and ordering
- Rating patterns (speed, scores given)
- Delivery feedback & Support interactions
- Location patterns throughout day

Is this creepy? ...A little bit. But this is how personalization works.

The Scale of Zomato's Data Infrastructure

Time Series

2.2 B

Active monitoring
points

Peak Rate

17.8 M

Metrics per second (NYE)

Logs

60 TB

Generated in a single day

Kafka

450+ M

Messages per minute

Services

300+

Microservices coordinated

Data Scanned

2 PB

Weekly by Trino analytics

Source: Zomato Engineering Blog, AWS Case Study

No Human Will Ever Look at This Data

- **Too big** — 60 TB in a single day
- **Too fast** — 17.8 million metrics per second
- **Too complex** — 300+ microservices interacting

The Automated Flow:

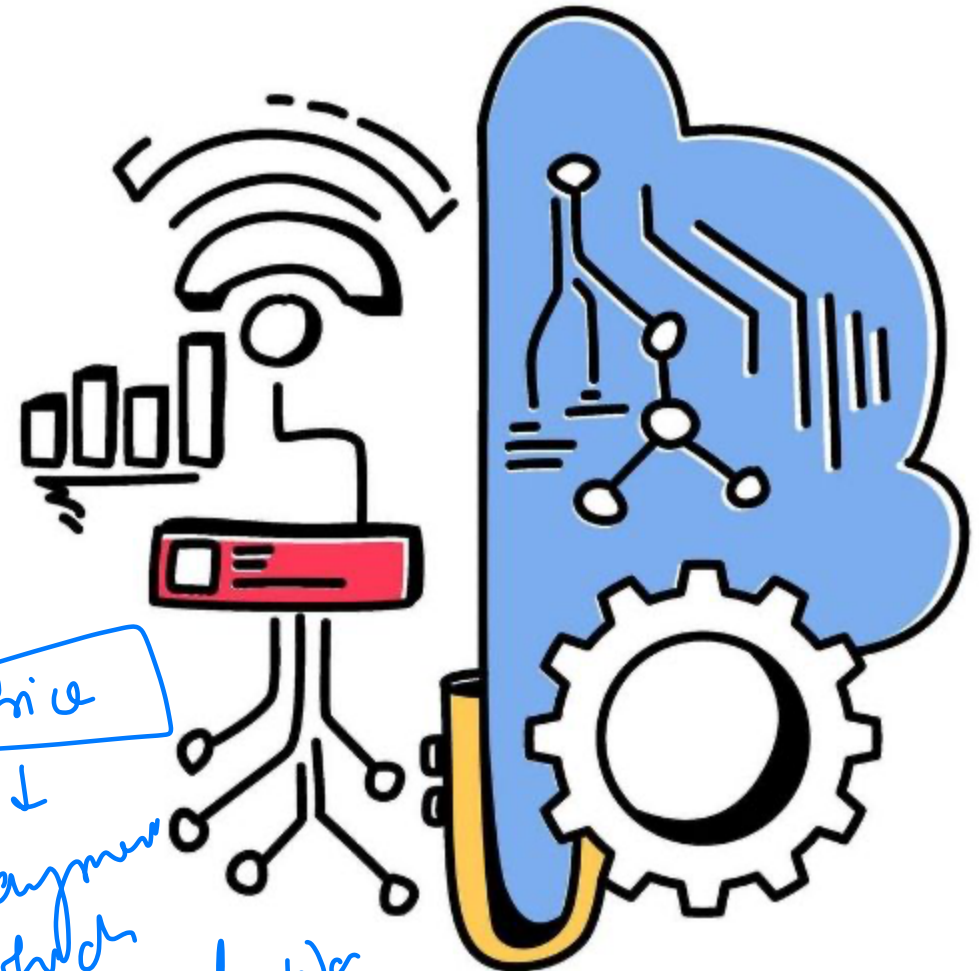
COLLECT → PROCESS → LEARN → DECIDE

"This is why data science skills matter. Modern businesses run on scales humans cannot comprehend."

ML 2
Time of ride
Rest → home

ML 2
Time
Loc X → Res

Price
↓
Payment
which
Card + Web



What You'll Learn to Build

Connecting Tonight's Systems to
Your 9-Month Journey



Everything We Discussed — You Will Learn to Build

Zomato System	What It Does	Course Topic	When
Personalization	Ranks restaurants for each user	Recommendation Systems	Month 7
ETA Prediction	Forecasts delivery time	Regression, Time Series	Month 5-6
Rider Matching	Assigns optimal rider	Optimization, OR	Month 6
Dynamic Pricing	Balances supply/demand	Economics + ML	Month 7
Data Infrastructure	Processes petabytes	Python, Pandas, SQL	Month 1-3

We start with Python and Data — because nothing works without clean data. Foundation first.

What Changed Tonight

Before Tonight, You saw:



- An app to order food
- "Just tap, wait, eat"
- Simple, magical interface

After Tonight, You see:

- 🧩 A personalization engine
- 🕒 A prediction system (LSTMs)
- 🧑‍🔧 An optimization network (Reinforcement Learning)
- ⚡ Real-time pricing models

Running 1,200 times per minute, in 500 milliseconds.

Thank You

Next Session

YouTube — The Algorithm Behind Your Addiction

Same approach. Different domain. Equally mind-bending.

"Next time you order food, you'll know exactly what's happening behind that screen."